

**ZERO-SHIFT SOURCE RIGIDITY AND KERNEL
ANNIHILATION
IN THE FINITE WEIL–CCM HIERARCHY**

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ABSTRACT. We continue the first-bad analysis of the canonical centered Weil–CCM hierarchy at the zero-shift endpoint, where predecessor blocks may be singular. The preceding first-bad theory reduces a hypothetical finite instability to a one-dimensional intrinsic cubic shell and proves, at every safe negative shift, an exact source-explicit cross-parity secular transfer. Accordingly, the negative-shift transfer cannot simply be specialized by substituting $\lambda = 0$.

We instead work directly with predecessor kernels and explicit zero-shift preimages. Let H_+ and H_- denote the intrinsic predecessor blocks in the even and odd reversal parities, let D denote centered-index parity transport, and let $\Phi_{N+1,L}$ denote the quadratic-normal functional of the actual canonical finite source matrix. For every $z \in \text{Ker } H_+$ we prove the denominator-free transport law

$$H_- Dz = \beta(z)d + \Phi_{N+1,L}(z)a,$$

where d and a are the two canonical odd predecessor correction directions and $\beta(z)$ is the normalized even shell-coupling coefficient. Projection onto the full odd predecessor kernel gives the exact compatibility identity

$$\beta(z) \text{Proj}_{\text{Ker } H_-} d + \Phi_{N+1,L}(z) \text{Proj}_{\text{Ker } H_-} a = 0.$$

When both canonical cubic couplings admit zero-shift predecessor preimages, the two shell responses satisfy the direct zero-shift transfer

$$\sigma_- = \alpha_0 \sigma_+ + \Gamma_0 \Phi_{N+1,L}(u_+),$$

proved at zero itself rather than by a resolvent limit. Under the same hypotheses we obtain the source-rigidity law

$$\boxed{\Gamma_0 \Phi_{N+1,L}(z) = 0 \quad (z \in \text{Ker } H_+)}.$$

Consequently, either the zero-shift cross-parity transfer coefficient degenerates or the exact production source functional annihilates the entire even predecessor kernel. In the nondegenerate branch this becomes an explicit pole–archimedean–prime cancellation law on every predecessor-kernel vector.

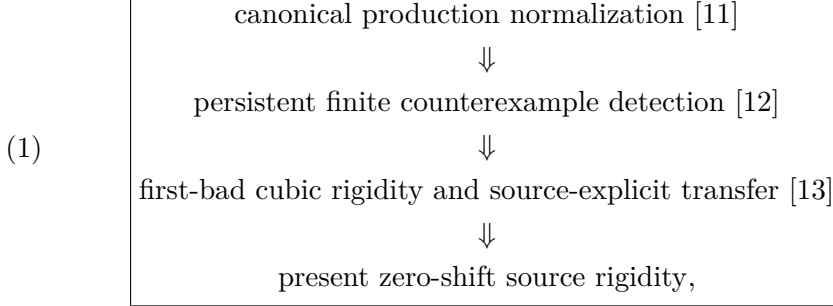
The results are finite rigidity statements. They do not prove zero-shift solvability at every first-bad state, do not prove $\Gamma_0 \neq 0$, do not exclude either branch of the resulting dichotomy, do not exclude the forced negative secular root, and do not prove the Riemann hypothesis. The theorem-bearing core is formally verified in Lean 4.

1. INTRODUCTION

Weil’s explicit formula admits a positivity formulation equivalent to the Riemann hypothesis. Finite and localized shadows of this positivity problem have been studied through Hermitian forms, variational principles, structured finite matrices, spectral constructions, and finite Guinand–Weil dictionaries; see Weil [16], Yoshida [17], Bombieri [2, 3], Connes–Consani [4], Connes–Consani–Moscovici [5], Connes–van Suijlekom [6], Groskin [7], and Suzuki [15].

The present paper belongs to a sequence of finite structural reductions. The first companion manuscript isolates a closed-form negative witness for a visible off-critical pair block [8]. The second compresses complete off-critical-zero detection to two translates and an explicit countable detector bank [9]. The third shows that the extension of a smooth finite Weil dictionary is governed by a single dimension-independent scalar defect [10]. The fourth reconciles the canonical finite source matrix with a normalization inconsistency in the displayed finite CCM formulas [11]. The fifth proves that one arbitrary hypothetical off-critical zero forces a genuine finite canonical negative vector and that the same physical vector remains negative at every larger centered size [12]. The sixth studies the first finite failure and derives a one-dimensional cubic secular architecture with an exact source-explicit cross-parity transfer [13].

Program lineage and direct dependencies. The historical chain and the logical dependencies should be distinguished. The canonical production matrix used below is the normalized object fixed in [11], and the immediate mathematical parent is the predecessor-shell, cubic-defect, and cross-parity theory of [13]. The persistent finite-detection theorem [12] supplies the upstream route from a hypothetical off-critical zero to the first-bad states studied in that parent paper, but it is *not* a hypothesis of the local zero-shift theorems proved here. The program lineage is therefore



while the direct imports used in the proofs below are the canonical normalization and the finite parity/cubic architecture of [11, 13]. The pair-local Gram witness, two-translate detector, and scalar-defect reduction remain relevant companion results, but they are not independent hypotheses here.

The present paper begins after the first-bad rigidity problem. In [13], global minimality at a fixed aperture forces both predecessor reversal-parity sectors to be nonnegative, while a bad successor parity contains exactly one new intrinsic degree of freedom. A canonical cubic generator meets that one-dimensional shell, reducing the negative-shift eigenvalue problem to an exact scalar secular equation.

At a common safe negative shift $\lambda < 0$, the same work identifies the surviving cubic parity-defect coefficient with a quadratic-normal moment of the actual canonical source matrix and derives the source-explicit transfer

$$(2) \quad F_-(\lambda) = \alpha(\lambda)F_+(\lambda) + \Gamma(\lambda)\Phi_{N+1,L}(u_+(\lambda)).$$

It deliberately does not assert that α , Γ , the relevant overlap, or the source moment is nonzero, and it does not exclude either zero-shift branch.

The present paper asks what this architecture forces *exactly at zero shift*. The difficulty is structural. At $\lambda < 0$, predecessor nonnegativity makes the shifted predecessor operator invertible. At $\lambda = 0$, the predecessor itself may have nontrivial kernel, so an expression involving H^{-1} is unavailable and no continuity argument by itself licenses substitution into (2).

Our route is denominator-free. Let H_+ and H_- denote the intrinsic even and odd predecessor blocks, let

$$D : V_{N+1}^+ \longrightarrow V_{N+1}^-$$

be the centered-index parity map, whose restriction sends the even predecessor carrier W_+ into the odd predecessor carrier W_- . We use the same symbol D for this restriction. Let c_+ and c_- be the canonical cubic shell vectors. The transported even shell and the complete odd cubic generator have predecessor corrections $d, a \in W_-$, so schematically

$$(3) \quad Dc_+ = d + c_-, \quad g_- = a + c_-.$$

For $z \in W_+$ let $\beta(z)$ denote the normalized coupling to the even cubic shell.

The first new transport identity is

$$(4) \quad \boxed{H_- Dz = \beta(z)d + \Phi_{N+1,L}(z)a, \quad z \in \text{Ker } H_+}.$$

Thus an even predecessor null direction does not cross parity arbitrarily: its odd predecessor forcing is confined to two canonical directions, and the coefficient of one of them is the exact production source moment.

Projecting (4) into the entire odd predecessor kernel gives

$$(5) \quad \boxed{\beta(z) \text{Proj}_{\text{Ker } H_-} d + \Phi_{N+1,L}(z) \text{Proj}_{\text{Ker } H_-} a = 0}.$$

No one-dimensional-kernel hypothesis is used.

We then suppose that the two cubic couplings admit explicit zero-shift preimages

$$(6) \quad H_+ x_+ = b_+, \quad H_- x_- = b_-.$$

Writing

$$(7) \quad u_+ = c_+ - x_+, \quad u_- = c_- - x_-,$$

one obtains the direct zero-shift transfer

$$(8) \quad \boxed{\sigma_- = \alpha_0 \sigma_+ + \Gamma_0 \Phi_{N+1,L}(u_+)}.$$

This identity is proved at zero from the explicit predecessor equations (6); no Laurent expansion, Moore–Penrose pseudoinverse, or limit of the negative-shift resolvent is used.

The main rigidity appears when (4) is combined with both zero-shift preimages. Symmetry of H_+ forces $\beta(z) = 0$ on $\text{Ker } H_+$, and the odd

transport becomes purely source-driven. Pairing against x_- then gives

$$(9) \quad \boxed{\Gamma_0 \Phi_{N+1,L}(z) = 0 \quad (z \in \text{Ker } H_+).}$$

Hence

$$(10) \quad \Gamma_0 \neq 0 \implies \Phi_{N+1,L}|_{\text{Ker } H_+} = 0.$$

Conversely, a single source-visible even predecessor null direction forces $\Gamma_0 = 0$.

The coefficient Γ_0 has the exact overlap representation

$$(11) \quad \Gamma_0 = \frac{\langle u_-, g_- \rangle}{\langle c_-, c_- \rangle}.$$

Because $c_- \neq 0$, the exceptional coefficient branch is exactly

$$(12) \quad \Gamma_0 = 0 \iff \langle u_-, g_- \rangle = 0.$$

Thus, under the two preimage hypotheses, the zero-shift kernel geometry is compressed into the structural alternative

$$(13) \quad \boxed{\Phi_{N+1,L}|_{\text{Ker } H_+} = 0 \quad \text{or} \quad u_- \perp g_-}.$$

The notation in (13) is deliberately shorthand for the precise implications proved below; no claim is made that one of the branches is impossible.

Main contributions. The paper establishes the following finite theorem suite.

- (i) An exact production decomposition of the quadratic-normal source functional into one surviving pole profile, reduced archimedean diagonal and off-diagonal moments, and a finite von Mangoldt-weighted source-atom sum.
- (ii) A denominator-free cross-parity transport theorem for every even predecessor-kernel vector,

$$H_- Dz = \beta(z)d + \Phi(z)a.$$

- (iii) An exact compatibility identity obtained by projecting that transport into the entire odd predecessor kernel.
- (iv) A direct zero-shift cross-parity shell-response theorem constructed from explicit predecessor preimages rather than from a resolvent limit.
- (v) A metric interpretation of Γ_0 as the normalized overlap between the canonical odd zero-shift trial and the complete odd cubic generator.
- (vi) The source-rigidity product law

$$\Gamma_0 \Phi|_{\text{Ker } H_+} = 0.$$

- (vii) The resulting nondegenerate kernel-annihilation law and its exact pole–archimedean–prime cancellation form.

Claim boundary. Nothing in this paper proves the Riemann hypothesis. We do not prove that both cubic couplings admit zero-shift preimages at every globally first-bad state; do not prove that Γ_0 is nonzero; do not prove that Φ is nonzero on a predecessor kernel; do not exclude the orthogonality branch $\Gamma_0 = 0$; do not exclude the source-annihilation branch; do not prove universal nonnegativity of the finite canonical matrices; and do not exclude the forced negative secular root of [13]. No inverse or pseudoinverse is implicitly introduced at zero shift, and no limit relation between (α_0, Γ_0) and the negative-shift coefficients $(\alpha(\lambda), \Gamma(\lambda))$ is asserted. The main product theorem is local in L and N ; it does not itself assume first-badness or the existence of an off-critical zero.

2. CANONICAL CENTERED HIERARCHY AND IMPORTED FIRST-BAD STRUCTURE

We use the canonical source normalization established in [11] and the boundary-flat parity/cubic hierarchy developed in [13]. The earlier persistent-detection paper [12] supplies the upstream counterexample route but is not needed as a hypothesis in the local algebra below. The analytic construction of the finite matrix is not repeated here.

Fix $L > 0$. For $K \in \mathbb{N}$, write

$$A_{K,L} \in M_{2K+1}(\mathbb{C})$$

for the canonical direct-source finite Weil-CCM matrix, indexed by the centered frequencies $-K, \dots, K$. For a coefficient vector $u = (u_{-K}, \dots, u_K)$ define the centered moments

$$(14) \quad M_j^{(K)}(u) = \sum_{n=-K}^K n^j u_n, \quad j \geq 0,$$

and the boundary-flat sector

$$(15) \quad V_K = \{u \in \mathbb{C}^{2K+1} : M_0^{(K)}(u) = M_1^{(K)}(u) = M_2^{(K)}(u) = 0\}.$$

The canonical quadratic form is

$$(16) \quad Q_{K,L}^{\text{can}}(u) = u^* A_{K,L} u.$$

On V_K it agrees with the complete Weil self-form of the represented hard-window finite test in the normalization of [11, 12].

Let J_K denote centered reversal,

$$(17) \quad (J_K u)_n = u_{-n}.$$

The canonical matrix commutes with reversal, so

$$(18) \quad V_K = V_K^+ \oplus V_K^-,$$

where $V_K^\pm$ are the reversal-even and reversal-odd sectors. Let $T_{\pm,K,L}$ denote the corresponding compressed self-adjoint operators.

At a one-step transition from predecessor size N to successor size $N + 1$, the first-bad theory of [13] supplies the orthogonal decomposition

$$(19) \quad V_{N+1}^p = W_{p,N} \oplus S_{p,N}, \quad p \in \{+, -\},$$

where $W_{p,N}$ is the inherited predecessor carrier and $S_{p,N}$ is a one-dimensional intrinsic shell. Let $c_{p,N} \neq 0$ be the canonical cubic shell generator. We suppress N, L when no confusion can arise and write

$$W_{\pm} := W_{\pm,N}, \quad c_{\pm} := c_{\pm,N}.$$

Following [13], we deliberately use H rather than A for the predecessor block, since $A_{K,L}$ already denotes the canonical source matrix. Define

$$(20) \quad H_{\pm} := P_{W_{\pm}} T_{\pm,N+1,L} |_{W_{\pm}} : W_{\pm} \rightarrow W_{\pm}$$

and the shell-to-predecessor couplings

$$(21) \quad b_{\pm} := P_{W_{\pm}} T_{\pm,N+1,L} c_{\pm} \in W_{\pm}.$$

The predecessor blocks are self-adjoint in the induced inner products. Throughout, $\langle \cdot, \cdot \rangle$ is conjugate-linear in the first slot and linear in the second, matching the convention used in the formalization and in [13].

The centered-index parity map is an algebraic linear map

$$(22) \quad D : V_{N+1}^+ \longrightarrow V_{N+1}^-.$$

It sends W_+ into W_- , and we use the same symbol D for the induced restriction $W_+ \rightarrow W_-$. No isometry or unitary property of D is assumed. The transported even cubic shell and the complete odd cubic generator decompose as

$$(23) \quad Dc_+ = d + c_-,$$

$$(24) \quad g_- = a + c_-,$$

with canonical predecessor corrections $d, a \in W_-$. These are the paper-level counterparts of the formal objects `oddIndexCubicShellPredecessorPart` and `oddCubicGeneratorPredecessorPart`.

For $w \in W_p$, define the normalized cubic shell-coupling coefficient

$$(25) \quad \beta_p(w) := \frac{\langle b_p, w \rangle}{\langle c_p, c_p \rangle}.$$

In the even sector we abbreviate $\beta(w) := \beta_+(w)$. The denominator is nonzero because $c_p \neq 0$.

Remark 2.1 (Imported theorem package). The globally first-bad reduction, one-dimensional intrinsic shell, cubic defect factorization, negative-shift secular scalar, zero-shift kernel-range decomposition, and negative-shift source-explicit cross-parity transfer are imported from [13]. The present paper does not reprove those statements.

3. THE EXACT PRODUCTION SOURCE FUNCTIONAL

The first-bad theory identifies the even-to-odd cubic defect coefficient used there with a quadratic-normal moment of the canonical source matrix. The subsequent formal source decomposition makes that moment completely production-explicit.

For $K \geq 2$, let $n_K^{(2)}$ denote the centered quadratic normal used in the even boundary-flat sector. For a matrix M and an even boundary-flat vector v , write

$$\text{Mom}_K(M; v)$$

for the normalized matrix moment against $n_K^{(2)}$. At the formal level this is `quadraticNormalMatrixMoment`.

Define the exact canonical source functional by

$$(26) \quad \Phi_{K,L}(v) := \Phi_{K,L}^{\text{can}}(v),$$

where $\Phi_{K,L}^{\text{can}}$ is the formal `explicitCanonicalSourceMoment`. Its prime-source weight and coordinate are

$$(27) \quad \omega_q = \frac{\Lambda(q)}{\sqrt{q}}, \quad \xi_q(L) = 1 - \frac{\log q}{L}.$$

For the pole channel set

$$(28) \quad \delta_n(L) = L^2 + 16\pi^2 n^2, \quad p_n(L) = \frac{L}{\delta_n(L)},$$

and

$$(29) \quad s_{\text{pole}}(L) = 32L \sinh^2(L/4).$$

Let $p_{K,L}$ be the centered vector with entries $p_n(L)$. The exact source functional is

$$(30) \quad \begin{aligned} \Phi_{K,L}(v) = & \frac{s_{\text{pole}}(L) (p_{K,L} \cdot v) (\overline{p_{K,L} \cdot n_K^{(2)}})}{\langle n_K^{(2)}, n_K^{(2)} \rangle} \\ & - \text{Mom}_K(A_{K,L}^{\text{arch,diag}}; v) - \text{Mom}_K(A_{K,L}^{\text{arch,off}}; v) \\ & - \sum_{q=2}^{\lfloor e^L \rfloor} \frac{\Lambda(q)}{\sqrt{q}} \text{Mom}_K(S(\xi_q(L)); v). \end{aligned}$$

Here $A_{K,L}^{\text{arch,diag}}$ and $A_{K,L}^{\text{arch,off}}$ are the reduced canonical archimedean diagonal and off-diagonal matrices, $S(\xi)$ is the elementary source matrix, and $\cdot$ denotes the bilinear coefficient dot product used in the rank-one source factorization (not the Hermitian inner product). Equation (30) is simply paper notation for the formal definition of `explicitCanonicalSourceMoment`.

The index-independent canonical archimedean scalar correction does not appear in (30). It has not been discarded from the canonical matrix. Rather, this particular quadratic-normal observable kills the scalar channel because the quadratic normal is orthogonal to the even boundary-flat sector. The

scalar correction remains present in scalar-sensitive absolute energy calculations outside the present theorem.

Theorem 3.1 (Exact production source decomposition). *Let $L > 0$, $K \geq 2$, and $v \in V_K^+$. Then the quadratic-normal source moment of the canonical production matrix equals exactly the expression $\Phi_{K,L}(v)$ in (30). Moreover, the cubic parity-defect coefficient of the first-bad theory is equal to the same production functional $\Phi_{K,L}(v)$.*

Proof. The canonical source matrix splits exactly into pole minus archimedean minus prime channels. Linearity of the quadratic-normal matrix moment reduces the calculation channel by channel. The reduced archimedean decomposition removes the scalar identity channel from this observable; the pole channel factors into even and odd rank-one profiles, with the odd profile vanishing against even boundary-flat input; and the prime channel expands into the finite source-matrix sum with weight $\Lambda(q)/\sqrt{q}$ and coordinate $1 - \log q/L$. The second assertion is the source-explicit identification of the cubic defect coefficient. In Lean these statements are `evenQuadraticSourceMoment_eq_explicitCanonicalSourceMoment` and `cubicDefectFunctional_eq_explicitCanonicalSourceMoment`. $\square$

Remark 3.2. Theorem 3.1 is a bridge, not the main novelty claim of this paper. Its role is to ensure that the scalar Φ entering the zero-shift geometry below is the literal production pole–archimedean–prime observable, not an abstract defect symbol.

4. DENOMINATOR-FREE CROSS-PARITY KERNEL TRANSPORT

Fix $L > 0$, $N \geq 1$, and let $z \in W_+$ satisfy

$$(31) \quad H_+ z = 0.$$

Because z lies in the predecessor carrier, the predecessor component of $T_+ z$ vanishes. The complementary successor shell is one-dimensional, so the full image is a scalar multiple of the canonical cubic shell.

Lemma 4.1 (Pure shell response of an even predecessor-kernel vector). *If $z \in \text{Ker } H_+$, then*

$$(32) \quad T_+ z = \beta(z) c_+.$$

Proof. The predecessor projection of $T_+ z$ is $H_+ z = 0$. Hence $T_+ z$ belongs to the one-dimensional shell $S_+ = \mathbb{C} c_+$. Its canonical quotient coordinate is the normalized shell-coupling coefficient $\beta(z)$ from (25). $\square$

The centered-index intertwining defect from [13], combined with Theorem 3.1, gives

$$(33) \quad T_- D z = D(T_+ z) + \Phi_{N+1,L}(z) g_-.$$

Using (23), (24), and Lemma 4.1,

$$(34) \quad \begin{aligned} T_- Dz &= \beta(z)(d + c_-) + \Phi(z)(a + c_-) \\ &= (\beta(z)d + \Phi(z)a) + (\beta(z) + \Phi(z))c_-. \end{aligned}$$

This is the full successor-carrier transport identity.

Theorem 4.2 (Denominator-free kernel/source transport). *Let $L > 0$, $N \geq 1$, and $z \in W_+$ satisfy $H_+ z = 0$. Then*

$$(35) \quad \boxed{H_- Dz = \beta(z)d + \Phi_{N+1,L}(z)a.}$$

The complementary cubic quotient is

$$(36) \quad \boxed{\text{quot}_-(T_- Dz) = \beta(z) + \Phi_{N+1,L}(z).}$$

Proof. Equation (34) is already decomposed into predecessor and shell coordinates. Projection onto W_- kills c_- and gives (35). The normalized shell quotient kills the predecessor component and sends c_- to 1, giving (36). $\square$

The formal predecessor and shell statements are, respectively,

`oddPredecessorBlock_evenIndex_eq_beta_add_source_of_evenKernel`
`oddCubicCouplingQuotient_evenIndex_eq_beta_add_source_of_evenKernel`

Remark 4.3 (Firewalls). Theorem 4.2 uses no inverse or pseudoinverse of either predecessor block, no Laurent expansion, no resolvent limit, no assumption on the dimension of either kernel, no isometry of D , and no sign or nonzeroness assertion for β or Φ .

Because H_- is self-adjoint,

$$(37) \quad \text{Ran } H_- \perp \text{Ker } H_-.$$

Let

$$K_- : W_- \longrightarrow \text{Ker } H_-$$

denote orthogonal projection.

Corollary 4.4 (Whole-kernel projected compatibility). *Under the hypotheses of Theorem 4.2,*

$$(38) \quad \boxed{\beta(z)K_- d + \Phi_{N+1,L}(z)K_- a = 0.}$$

Proof. Apply K_- to (35). Since $H_- Dz \in \text{Ran } H_-$, its projection into $\text{Ker } H_-$ is zero. $\square$

Equation (38) is a vector identity in the full odd predecessor kernel; no one-dimensional-kernel reduction is made. The formal development also specializes it to the actual even cubic-coupling kernel component.

5. EXPLICIT ZERO-SHIFT PREIMAGES

The safe negative-shift theory uses $H_p - \lambda I$ with $\lambda < 0$, which is invertible under predecessor nonnegativity. Zero shift is different. We therefore make no definition involving H_p^{-1} .

Assume instead that the cubic couplings belong to the ranges of the two predecessor blocks. Choose explicit vectors

$$x_+ \in W_+, \quad x_- \in W_-$$

satisfying

$$(39) \quad H_+ x_+ = b_+,$$

$$(40) \quad H_- x_- = b_-.$$

Define the canonical zero-shift trial vectors

$$(41) \quad u_+ = c_+ - x_+, \quad u_- = c_- - x_-.$$

Since predecessor and shell components are orthogonal,

$$(42) \quad \langle x_{\pm}, c_{\pm} \rangle = 0.$$

The cubic quotient coordinate of each $u_{\pm}$ is therefore 1.

Equations (39)–(40) cancel the predecessor components of the corresponding successor images. Hence there exist unique shell-response scalars $\sigma_{\pm} \in \mathbb{C}$ such that

$$(43) \quad T_{\pm} u_{\pm} = \sigma_{\pm} c_{\pm}.$$

At the formal level these are the values of

`cubicZeroShiftShellResponseScalar`.

For the selected odd preimage x_- define

$$(44) \quad \alpha_0 := 1 - \frac{\langle x_-, d \rangle}{\langle c_-, c_- \rangle},$$

$$(45) \quad \Gamma_0 := 1 - \frac{\langle x_-, a \rangle}{\langle c_-, c_- \rangle}.$$

These are the paper-level forms of `crossParityZeroShiftAlpha` and `crossParityZeroShiftGamma`.

Remark 5.1 (Preimage dependence). The coefficients α_0 and Γ_0 are attached to the selected odd preimage x_- . When H_- is singular, the paper does not assert that the coefficients are individually independent of the choice of preimage. The product-rigidity theorem below is stated for every selected pair of preimages satisfying (39)–(40).

Remark 5.2 (No limit assertion). We do not assert

$$\alpha_0 = \lim_{\lambda \uparrow 0} \alpha(\lambda), \quad \Gamma_0 = \lim_{\lambda \uparrow 0} \Gamma(\lambda).$$

No such statement is used.

6. DIRECT SOURCE-EXPLICIT TRANSFER AT ZERO SHIFT

The centered-index transport of the even zero-shift trial has the canonical odd shell component. Combining (43) in the even parity with the cubic defect factorization and the source identification of Theorem 3.1 gives the full transported response

$$(46) \quad \begin{aligned} T_- Du_+ &= (\sigma_+ d + \Phi_{N+1,L}(u_+)a) \\ &\quad + (\sigma_+ + \Phi_{N+1,L}(u_+))c_-. \end{aligned}$$

This is the paper form of `parityCompressedCanonical_odd_evenIndex_zeroShiftTrial_eq`.

Since both Du_+ and u_- have shell quotient coordinate 1,

$$(47) \quad w := Du_+ - u_-$$

lies in the odd predecessor carrier W_- . Subtracting the response $T_- u_- = \sigma_- c_-$ from (46) yields

$$(48) \quad H_- w = \sigma_+ d + \Phi_{N+1,L}(u_+)a,$$

while the shell quotient of $T_- w$ is

$$(49) \quad \sigma_+ + \Phi_{N+1,L}(u_+) - \sigma_-.$$

For any predecessor vector w , the shell quotient of $T_- w$ is the normalized coupling

$$(50) \quad \frac{\langle b_-, w \rangle}{\langle c_-, c_- \rangle}.$$

Using $b_- = H_- x_-$ and self-adjointness,

$$(51) \quad \langle b_-, w \rangle = \langle H_- x_-, w \rangle = \langle x_-, H_- w \rangle.$$

Substitution of (48) gives

$$(52) \quad \langle b_-, w \rangle = \sigma_+ \langle x_-, d \rangle + \Phi_{N+1,L}(u_+) \langle x_-, a \rangle.$$

Equations (44)–(52) give the direct endpoint transfer.

Theorem 6.1 (Direct zero-shift cross-parity transfer). *Let $L > 0$, $N \geq 1$, and suppose x_+ and x_- satisfy (39)–(40). Then*

$$(53) \quad \boxed{\sigma_- = \alpha_0 \sigma_+ + \Gamma_0 \Phi_{N+1,L}(u_+).}$$

Proof. By (49)–(52),

$$\sigma_+ + \Phi(u_+) - \sigma_- = \frac{\sigma_+ \langle x_-, d \rangle + \Phi(u_+) \langle x_-, a \rangle}{\langle c_-, c_- \rangle}.$$

Rearranging and using (44)–(45) yields (53). $\square$

The exact formal theorem is `cubicZeroShiftShellResponseScalar_crossParity_explicitSource_transfer`.

Remark 6.2. Theorem 6.1 is proved at $\lambda = 0$ itself. It is not presented as a continuity specialization of the safe negative-shift transfer (2).

7. ZERO-SHIFT SOURCE RIGIDITY

We now derive the principal result. Let $z \in \text{Ker } H_+$ and assume the even zero-shift preimage (39) exists. Since H_+ is self-adjoint,

$$(54) \quad \langle b_+, z \rangle = \langle H_+ x_+, z \rangle = \langle x_+, H_+ z \rangle = 0.$$

Hence

$$(55) \quad \boxed{\beta(z) = 0.}$$

Theorem 4.2 collapses to the one-channel source law

$$(56) \quad \boxed{H_- Dz = \Phi_{N+1,L}(z)a,}$$

and its shell quotient becomes

$$(57) \quad \text{quot}_-(T_- Dz) = \Phi_{N+1,L}(z).$$

Now suppose the odd zero-shift preimage (40) also exists and write

$$\mu := \Phi_{N+1,L}(z).$$

Equation (57), expressed through the shell-coupling metric, gives

$$(58) \quad \mu = \frac{\langle b_-, Dz \rangle}{\langle c_-, c_- \rangle}.$$

Using $b_- = H_- x_-$ and self-adjointness,

$$(59) \quad \langle b_-, Dz \rangle = \langle x_-, H_- Dz \rangle = \mu \langle x_-, a \rangle,$$

where the last equality uses (56). Hence

$$(60) \quad \mu = \mu \frac{\langle x_-, a \rangle}{\langle c_-, c_- \rangle}.$$

Rearranging and using (45) gives the product law.

Theorem 7.1 (Zero-shift source-rigidity product law). *Let $L > 0$ and $N \geq 1$. Suppose there exist $x_+ \in W_+$ and $x_- \in W_-$ satisfying*

$$H_+ x_+ = b_+, \quad H_- x_- = b_-.$$

Then every $z \in \text{Ker } H_+$ satisfies

$$(61) \quad \boxed{\Gamma_0 \Phi_{N+1,L}(z) = 0.}$$

Proof. Equations (54)–(60) show

$$\left(1 - \frac{\langle x_-, a \rangle}{\langle c_-, c_- \rangle}\right) \Phi_{N+1,L}(z) = 0.$$

The factor in parentheses is Γ_0 by (45). □

The exact formal theorem is `crossParityZeroShiftGamma_mul_explicitSource_eq_zero_on_evenKernel`. Its proof is deliberately product-valued: no division by Γ_0 or by the source moment is hidden.

Corollary 7.2 (Nondegenerate kernel annihilation). *Under the hypotheses of Theorem 7.1, if $\Gamma_0 \neq 0$, then*

$$(62) \quad \boxed{\Phi_{N+1,L}(z) = 0 \quad \text{for every } z \in \text{Ker } H_+}.$$

Equivalently,

$$(63) \quad \boxed{\Phi_{N+1,L}|_{\text{Ker } H_+} = 0}.$$

Proof. Apply Theorem 7.1 to an arbitrary kernel vector and cancel the nonzero factor Γ_0 . $\square$

The formal corollary is `explicitCanonicalSourceMoment_eq_zero_on_evenKernel_of_zeroShiftGamma_ne_zero`.

Corollary 7.3 (Source-visible kernel forces transfer degeneration). *Under the hypotheses of Theorem 7.1, if there exists $z \in \text{Ker } H_+$ with*

$$\Phi_{N+1,L}(z) \neq 0,$$

then

$$(64) \quad \boxed{\Gamma_0 = 0}.$$

Proof. Apply (61) to the chosen vector and cancel its nonzero source value. $\square$

Thus Theorem 7.1 compresses the zero-shift singular branch into the precise alternative

$$(65) \quad \boxed{\Gamma_0 = 0 \quad \text{or} \quad \Phi_{N+1,L}|_{\text{Ker } H_+} = 0}.$$

No conclusion is presently drawn about which branch occurs at a forced canonical first-bad state.

8. GEOMETRIC INTERPRETATION OF THE EXCEPTIONAL BRANCH

The coefficient Γ_0 is not merely an algebraic placeholder. Recall $u_- = c_- - x_-$ and $g_- = a + c_-$. Predecessor-shell orthogonality gives

$$(66) \quad \langle x_-, c_- \rangle = 0, \quad \langle c_-, a \rangle = 0.$$

Therefore

$$(67) \quad \begin{aligned} \langle u_-, g_- \rangle &= \langle c_- - x_-, a + c_- \rangle \\ &= \langle c_-, c_- \rangle - \langle x_-, a \rangle. \end{aligned}$$

Using (45),

$$(68) \quad \boxed{\Gamma_0 = \frac{\langle u_-, g_- \rangle}{\langle c_-, c_- \rangle}}.$$

This is the paper-level form of `crossParityZeroShiftGamma_eq_trial_cubic_overlap_div`.

Corollary 8.1 (Geometric form of zero-shift degeneration). *For every selected odd zero-shift preimage x_- ,*

$$(69) \quad \boxed{\Gamma_0 = 0 \iff \langle u_-, g_- \rangle = 0.}$$

Proof. The denominator in (68) is nonzero because $c_- \neq 0$. $\square$

Corollary 8.2 (Source/kernel versus cubic-overlap dichotomy). *Under the two zero-shift solvability hypotheses,*

$$(70) \quad \boxed{\Phi_{N+1,L}|_{\text{Ker } H_+} = 0 \quad \text{or} \quad u_- \perp g_-}.$$

More sharply, if there exists a source-visible vector $z \in \text{Ker } H_+$, then necessarily $u_- \perp g_-$.

Proof. Combine (65) with Corollary 8.1. $\square$

The converse implication is not asserted: $\Gamma_0 = 0$ does not imply that Φ is nonzero on the even predecessor kernel.

9. ARITHMETIC KERNEL-BALANCE LAW

Corollary 7.2 has a concrete arithmetic meaning because Φ is the exact production source functional of Theorem 3.1. Suppose the two zero-shift preimages exist and $\Gamma_0 \neq 0$. Let $z \in \text{Ker } H_+$. Then $\Phi_{N+1,L}(z) = 0$, and (30) gives

$$(71) \quad \begin{aligned} 0 = & \frac{s_{\text{pole}}(L) (p_{N+1,L} \cdot z) (\overline{p_{N+1,L} \cdot n_{N+1}^{(2)}})}{\langle n_{N+1}^{(2)}, n_{N+1}^{(2)} \rangle} \\ & - \text{Mom}_{N+1}(A_{N+1,L}^{\text{arch,diag}}; z) - \text{Mom}_{N+1}(A_{N+1,L}^{\text{arch,off}}; z) \\ & - \sum_{q=2}^{\lfloor e^L \rfloor} \frac{\Lambda(q)}{\sqrt{q}} \text{Mom}_{N+1}(S(1 - \log(q)/L); z). \end{aligned}$$

Corollary 9.1 (Arithmetic kernel-balance law). *Assume the two cubic couplings admit zero-shift preimages and $\Gamma_0 \neq 0$. Then every even predecessor null direction satisfies the exact production pole–archimedean–prime cancellation (71).*

Proof. By Corollary 7.2, $\Phi_{N+1,L}(z) = 0$ for every $z \in \text{Ker } H_+$. Expand Φ using Theorem 3.1. $\square$

Equation (71) is a cancellation statement, not a factorwise positivity theorem. No sign is asserted for the pole, archimedean, or source-atom contributions individually. Nor does the theorem assert that $\text{Ker } H_+$ is nontrivial: if H_+ is regular, the annihilation statement is vacuous.

The content is therefore specific to singular predecessor states. If a nontrivial even predecessor kernel survives simultaneously with two-sided zero-shift solvability and a nondegenerate odd transfer coefficient, the actual canonical arithmetic source must be invisible to that entire kernel.

10. RELATION TO THE PRECEDING FIRST-BAD SECULAR THEORY

The role of the present paper is clearest when compared directly with [13]. At a safe negative shift,

$$(72) \quad F_-(\lambda) = \alpha(\lambda)F_+(\lambda) + \Gamma(\lambda)\Phi_{N+1,L}(u_+(\lambda)), \quad \lambda < 0.$$

The predecessor equation is then uniquely solvable because $H_p - \lambda I$ is coercive and invertible.

The present paper does not merely append the endpoint $\lambda = 0$ to (72). Zero shift introduces a different geometric object: the predecessor kernel. The progression is

$$(73) \quad \begin{aligned} \lambda < 0 : & \text{ invertible predecessor} \longrightarrow \text{secular transfer,} \\ \lambda = 0 : & \text{ potentially singular kernel geometry} \longrightarrow \text{kernel/source transport} \\ & \longrightarrow \text{source rigidity.} \end{aligned}$$

In particular, Theorem 4.2 concerns an arbitrary vector of $\text{Ker } H_+$ rather than the selected secular trial. The product law (61) is additional rigidity at the potentially singular zero-shift endpoint; it is substantive when the even predecessor kernel is nontrivial.

The sequence of theorem stages in the present program can therefore be summarized as

$$(74) \quad \begin{array}{l} \text{Paper 5:} \quad \text{off-critical zero} \longrightarrow \text{persistent finite negativity,} \\ \text{Paper 6:} \quad \text{first finite failure} \longrightarrow \text{cubic secular rigidity,} \\ \text{Present:} \quad \text{zero-shift kernel geometry} \longrightarrow \text{source/kernel rigidity.} \end{array}$$

Here Paper 5 is [12] and Paper 6 is [13]; the canonical normalization of [11] underlies both and the present paper.

11. COMPARISON WITH NEARBY FINITE FRAMEWORKS

The novelty claim below is intentionally narrow. Table 1 records the nearest finite structures rather than presenting generic linear algebra as new. Suzuki [15] provides a recent unified operator-theoretic perspective on the Weil quadratic form, but does not state the kernel/source transport or product law considered here.

Table 1: Comparison with nearby finite frameworks.

Feature	Closest prior position	Present role
Finite negative geometry associated with off-critical data	Bombieri supplies finite negative-index phenomena in a different finite spectral model.	inherited input

Feature	Closest prior position	Present role
Exact finite dictionary input	Groskin gives an exact finite Guinand–Weil dictionary on the real-even Galerkin sector [7]; the canonical normalization used here is fixed in [11].	inherited/used
First-bad predecessor–shell architecture	Not the stated endpoint of Bombieri or Groskin; supplied here by [13].	inherited input
Denominator-free even-kernel to odd-predecessor transport	Not identified in the audited Weil/CCM sources in this formulation.	proved here
Production source moment as a transport coefficient	The source-moment identification begins in [13]; the exact production decomposition is made explicit here.	made explicit here
Direct zero-shift cross-parity transfer from explicit preimages	Not identified in the audited sources in this formulation.	proved here
Whole-kernel product law $\Gamma_0\Phi _{\text{Ker } H_+} = 0$	Not identified in the audited sources in this formulation.	headline result
Lean formalization of theorem core	Not part of the cited classical finite frameworks.	yes

The entries “not identified in audited sources” are deliberately weaker than universal priority claims. They report the outcome of the targeted literature comparison used for this manuscript.

12. FORMAL VERIFICATION IN LEAN 4

For broader background on formalizing zeta and L -functions in Lean, see [14]. The theorem core of the present paper is formalized in the public repository

`HMarcusWH/zeta-23-lean`.

The source-explicit decomposition is carried by

`Zeta23/CCM/SourceMomentDecomposition.lean`,

the denominator-free kernel transport by

`Zeta23/CCM/KernelSourceTransport.lean`,

and the direct zero-shift transfer and product law by

`Zeta23/CCM/ZeroShiftCrossParityTransfer.lean`.

The proof-bearing theorem head for PR #134 is

(75) `753ee53a7fc08bd3be9a5a0f37417629122395f9,`

with Git tree

(76) `c142efa141036331d139c532d06e7a976c5b50c2.`

The merged main commit is

(77) `7f1fec480d1ccbff04a456ab937accf7b23cc1af,`

and has the identical Git tree (76). The RHRC research and Lean workflow completed successfully as run #870 on the theorem head, and the independent Permansson v0.1.6 formal-verification workflow completed successfully as run #643 on the same head.

The toolchain is

`leanprover/lean4:v4.33.0-rc2,`

and Mathlib is pinned to

`51e6992efd06126df61a496bebf8f49482a4e129.`

Table 2: Paper-to-Lean theorem correspondence.

Paper result	Lean declaration
Exact production source decomposition	<code>evenQuadraticSourceMoment_eq_explicitCanonicalSourceMoment</code>
Cubic defect equals	<code>cubicDefectFunctional_eq</code>
production source moment	<code>explicitCanonicalSourceMoment</code>
Pure shell image of even predecessor-kernel vector	<code>parityCompressedCanonical_evenKernel_eq_beta_smul_cubic</code>
Full transported kernel response	<code>parityCompressedCanonical_odd_evenIndex_kernel_eq</code>
Denominator-free predecessor transport	<code>oddPredecessorBlock_evenIndex_eq_beta_add_source_of_evenKernel</code>
Shell quotient transport	<code>oddCubicCouplingQuotient_evenIndex_eq_beta_add_source_of_evenKernel</code>
Whole-kernel projected compatibility	<code>evenKernel_crossParity_projected_compatibility</code>
Canonical coupling-kernel specialization	<code>cubicCouplingKernelPart_crossParity_projected_compatibility</code>
Transported zero-shift trial response	<code>parityCompressedCanonical_odd_evenIndex_zeroShiftTrial_eq</code>
Odd coupling-kernel forcing	<code>oddCubicCouplingKernelPart_eq_evenZeroShiftResponse_add_source</code>
Zero-shift Γ_0 overlap formula	<code>crossParityZeroShiftGamma_eq_trial_cubic_overlap_div</code>

Paper result	Lean declaration
Direct zero-shift source transfer	<code>cubicZeroShiftShellResponseScalar_crossParity_explicitSource_transfer</code>
Source-rigidity product law	<code>crossParityZeroShiftGamma_mul_explicitSource_eq_zero_on_evenKernel</code>
Nonzero- Γ_0 kernel annihilation	<code>explicitCanonicalSourceMoment_eq_zero_on_evenKernel_of_zeroShiftGamma_ne_zero</code>

Several paper statements, such as Corollaries 8.1, 7.3, and 9.1, are direct mathematical consequences of named compiler-verified declarations rather than separately named Lean declarations. They are described as such instead of being presented as additional formal theorem objects.

The formal modules explicitly maintain the same claim firewalls used in the prose: no zero-shift inverse, pseudoinverse, Laurent expansion, or resolvent limit; no one-dimensional-kernel assumption; no division by α_0 , Γ_0 , overlap, or source moment in the primary product theorem; no source sign or nonzeroness claim; no branch exclusion; no negative-root exclusion; and no RH theorem.

13. PRIORITY BOUNDARY AND NOVELTY STATEMENT

The broad ingredients used in this paper are classical. Kernel-range decompositions of finite self-adjoint operators, orthogonal projection onto kernels, solvability of singular linear systems, Schur complements, secular equations, and finite parity decompositions are standard linear algebra and operator theory; classical pseudoinverse criteria already treat singular positive-semidefinite block geometry [1]. Weil positivity and finite negative-index phenomena associated with off-critical spectral data are established background [16, 17, 2, 3]. Structured finite Weil matrices and exact finite dictionary constructions likewise precede the present work [4, 5, 6, 7].

No novelty claim is made for these ingredients in isolation.

The preceding first-bad manuscript [13] already proves the one-dimensional intrinsic cubic shell, the one-channel even-to-odd compression defect, the exact negative-shift secular scalar, the zero-shift kernel-range branch decomposition, the identification of the cubic defect coefficient with a canonical source moment, and the safe-negative-shift transfer (72). Its novelty claim is likewise attached to the exact finite composition rather than to generic Schur or resolvent algebra.

The narrower claim of the present paper concerns the additional structure forced at zero shift. In a targeted audit of the closest publicly available materials used in this project, we did not identify a prior theorem combining all of the following:

- (a) denominator-free centered-index transport of every even predecessor-kernel vector into the two canonical odd predecessor correction channels;
- (b) identification of one of those transport coefficients with the exact production quadratic-normal source functional;
- (c) a direct zero-shift source-explicit cross-parity shell transfer constructed from explicit predecessor preimages rather than from a resolvent limit;
- (d) the whole-kernel product law

$$\Gamma_0 \Phi|_{\text{Ker } H_+} = 0;$$

- (e) the resulting nondegenerate arithmetic kernel-annihilation law.

We therefore describe the result conservatively as an *apparently new zero-shift source-rigidity and kernel-annihilation theorem for the canonical finite Weil-CCM hierarchy*. This is a “to our knowledge” statement, not a claim of exhaustive universal priority. In particular, no priority claim is attached to the generic singular-linear-algebra implications used inside the proof.

14. OPEN PROBLEMS

The product law separates several questions that remain entangled at the zero-shift endpoint.

P1. Forced two-parity zero-shift solvability. Determine whether a globally first-bad canonical state necessarily satisfies

$$b_+ \in \text{Ran } H_+, \quad b_- \in \text{Ran } H_-.$$

The present theorem assumes these range conditions. The preceding first-bad theory supplies a regular/resonant branch dichotomy but does not prove that the range-solvable branch occurs in both parities.

P2. Nondegeneracy of the zero-shift transfer coefficient. Determine whether $\Gamma_0 \neq 0$ is forced at a canonical first-bad state. By Corollary 8.1, this is equivalent to excluding the exact orthogonality $u_- \perp g_-$. Generic finite-dimensional arguments cannot exclude such orthogonality.

P3. Source visibility on a predecessor kernel. Determine whether a nonzero vector $z \in \text{Ker } H_+$ can satisfy $\Phi_{N+1,L}(z) = 0$ for the exact canonical source functional. Theorem 7.1 shows that this is not peripheral: on the nondegenerate branch the entire predecessor kernel is forced into the nullspace of Φ .

P4. Arithmetic incompatibility of the kernel-balance law. Use the exact decomposition (71) to determine whether a nontrivial predecessor null direction can satisfy the required pole–archimedean–prime cancellation at a first-bad canonical state. The target is the combined production cancellation, not separate sign control of each source atom.

P5. Reverse-parity transport. The present result is directionally even-to-odd, inheriting the orientation of the cubic parity defect established in [13]. Determine whether an independently proved odd-to-even zero-shift transport exists and whether the two product laws can be combined. No symmetric reverse theorem is inferred from notation alone.

P6. Relation to regular-aperture selection. Determine how the source-rigidity theorem interacts with aperture selections for which the relevant predecessor blocks are regular. At such apertures $\text{Ker } H_{\pm} = \{0\}$, so the kernel-annihilation statement is vacuous. A complementary question is whether the theorem constrains exceptional singular apertures, for example through source-visible resonances or vanishing cubic-overlap coefficients isolated by canonical arithmetic structure.

15. CONCLUSION

The preceding finite Weil–CCM analysis reduces a hypothetical first finite instability to a one-dimensional cubic shell and an exact source-explicit secular transfer at safe negative shifts. The present work studies the zero-shift endpoint, where nontrivial predecessor kernels may appear.

For every even predecessor null vector $z \in \text{Ker } H_+$, centered-index parity transport satisfies the exact denominator-free identity

$$H_- D z = \beta(z) d + \Phi_{N+1,L}(z) a.$$

When the canonical cubic couplings admit zero-shift preimages in both reversal parities, symmetry kills the shell coefficient $\beta(z)$ on the even kernel and forces

$$\boxed{\Gamma_0 \Phi_{N+1,L}(z) = 0 \quad (z \in \text{Ker } H_+).}$$

Thus the zero-shift kernel geometry is rigid under the stated preimage hypotheses: either the cross-parity cubic transfer degenerates through the exact orthogonality $u_- \perp g_-$, or the production arithmetic source functional annihilates the entire even predecessor kernel. On the nondegenerate branch this becomes the exact pole–archimedean–prime balance law (71).

Neither branch is excluded here. Determining whether either can occur at a forced canonical first-bad state remains open.

DATA AND CODE AVAILABILITY

No empirical dataset is required for the theorem. The Lean 4 formalization, theorem-control material, and validation infrastructure are publicly available at

<https://github.com/HMarcusWH/zeta-23-lean>.

The manuscript is pinned to the PR #134 proof-bearing theorem head and merge/tree identifiers listed in Section 12.

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The author declares no competing interests.

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REFERENCES

- [1] Arthur Albert, *Conditions for Positive and Nonnegative Definiteness in Terms of Pseudoinverses*, SIAM J. Appl. Math. **17** (1969), no. 2, 434–440. doi:10.1137/0117041.
- [2] Enrico Bombieri, *Remarks on Weil’s quadratic functional in the theory of prime numbers, I*, Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei Mat. Appl. **11** (2000), no. 3, 183–233.
- [3] Enrico Bombieri, *A variational approach to the explicit formula*, Comm. Pure Appl. Math. **56** (2003), no. 8, 1151–1164. doi:10.1002/cpa.10089.
- [4] Alain Connes and Caterina Consani, *Spectral triples and ζ -cycles*, L’Enseignement Mathématique **69** (2023), no. 1–2, 93–148. doi:10.4171/LEM/1049.
- [5] Alain Connes, Caterina Consani, and Henri Moscovici, *Zeta Spectral Triples*, 2025, arXiv:2511.22755 [math.OA].
- [6] Alain Connes and Walter D. van Suijlekom, *Quadratic Forms, Real Zeros and Echoes of the Spectral Action*, Commun. Math. Phys. **406** (2025), Article 312. doi:10.1007/s00220-025-05493-1.
- [7] Akiva Groskin, *A finite Guinand–Weil dictionary and archimedean tail order for the truncated Weil quadratic form*, 2026, arXiv:2607.02828 [math.NT].
- [8] Marcus Hermansson, *Explicit Gram Witnesses for Off-Critical Pair Blocks in Finite Weil Quadratic Forms*, manuscript, 25 August 2026. Companion theorem manuscript in the present series.
- [9] Marcus Hermansson, *Complete Two-Translate Detectors for Off-Critical Zeros in Weil’s Quadratic Form: Theorem Core*, manuscript, 28 August 2026. Companion theorem manuscript in the present series.
- [10] Marcus Hermansson, *A Scalar-Defect Reduction for Finite Weil–CCM Dictionaries*, manuscript, 30 August 2026. Companion theorem manuscript in the present series.
- [11] Marcus Hermansson, *Exact Normalization Reconciliation in the Finite Weil–CCM Matrix*, manuscript, 31 August 2026. Direct dependency: canonical production normalization used in the present paper.
- [12] Marcus Hermansson, *Persistent Finite Detection of Off-Critical Zeros in the Weil–CCM Hierarchy*, manuscript, 3 September 2026. Upstream program theorem: finite counterexample detection and centered-size persistence.
- [13] Marcus Hermansson, *First-Bad Cubic Rigidity and Source-Explicit Cross-Parity Secular Transfer in the Weil–CCM Hierarchy*, manuscript, 8 September 2026. Direct dependency and immediate parent theorem manuscript.
- [14] David Loeffler and Michael Stoll, *Formalizing zeta and L-functions in Lean*, Annals of Formalized Mathematics **1** (2025), 43–56. doi:10.46298/afm.15328.
- [15] Masatoshi Suzuki, *Weil’s quadratic form via the screw function*, 2026, arXiv:2606.09096 [math.NT].
- [16] André Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, Comm. Sémin. Math. Univ. Lund, Tome Supplémentaire (1952), 252–265.
- [17] Hiroyuki Yoshida, *On Hermitian Forms Attached to Zeta Functions*, in *Zeta Functions in Geometry*, Advanced Studies in Pure Mathematics, vol. 21, Kinokuniya, 1992, 281–325.

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